

Aymeric Fromherz

Curriculum Vitae

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Education

- 2017-2021 **PhD**, *Electrical and Computer Engineering*, Carnegie Mellon University
A Proof-Oriented Approach to Low-Level, High-Assurance Programming
coadvised by Bryan Parno and Corina Pasareanu
- 2014–2015, **Master (M.Sc.)**, *Computer Science*, Paris, Summa cum laude
- 2016–2017 MPRI (Master Parisien de Recherche en Informatique)
- 2014–2015 **Licence (B.Sc.)**, *Mathematics*, École Normale Supérieure, Paris
- 2013–2014 **Licence (B.Sc.)**, *Computer Science*, École Normale Supérieure, Paris, Summa cum laude
- 2011–2013 **Preparatory classes**, *MPSI/MP* (Maths-Physics)*, Lycée du Parc, Lyon
- 2011 **Baccalauréat (French high school diploma)**, *Scientific*, Cité Scolaire Internationale, Lyon, Summa cum laude

Professional appointments

- 2022-now **Inria Starting Faculty Position**, *Prosecco Team*, Inria, France
- 2021-2022 **Postdoctoral Researcher**, *Prosecco Team*, Inria, France
- Summer 2020 **Research Intern**, *Supervised by Nikhil Swamy*, Microsoft Research, Redmond, WA, USA
- Summer 2019 **Research Intern**, *Supervised by Nikhil Swamy*, Microsoft Research, Redmond, WA, USA
- Sept 2015–
June 2016 **Development and Formal Proof of a MicroKernel**, ProvenRun, Paris, France,
Engineer position

Honors and Awards

- 2022 **ACM SIGSAC Dissertation Award**
- 2022 **A.G. Milnes Award**, *Carnegie Mellon University*, Awarded to a graduating ECE Ph.D. student for the Ph.D. thesis work judged to be of the highest quality and which has had, or is likely to have, significant impact in his or her field.
- 2019 **Cylab Presidential Fellow**
- 2017 **Fondation Monahan Fellow**
- 2017 **Google Summer of Code**, Java Pathfinder, Mentor
- 2016 **Google Summer of Code**, Java Pathfinder, Student
- 2013 **Admission at the École Normale Supérieure, Paris**, National exam for computer science majors, ranked 13th
- 2011 **Abitur**, (German high school diploma)

Publications

CV2F*: Combining CryptoVerif and F* for Proving Security of Protocol Implementations
Bruno Blanchet, Aymeric Fromherz, Charlie Jacomme, Benjamin Lipp, Emmanuel Mera, *To Appear at the ACM Conference on Computer and Communications Security (CCS)*, 2026

Scylla: Translating an Applicative Subset of C to Safe Rust, Aymeric Fromherz, Jonathan Protzenko, *International Conference on Object-Oriented Programming, Systems, Languages, and Applications (OOPSLA)*, 2026

Project Everest: Perspectives from Developing Industrial-grade High-Assurance Software, Danel Ahman, Karthikeyan Bhargavan, Chris Brzuska, Barry Bond, Jay Bosamiya, Antoine Delignat-

Lavaud, Aymeric Fromherz, Cédric Fournet, Sydney Gibson, Chris Hawblitzel, Cătălin Hrițcu, Markulf Kohlweiss, Guido Martinez, Haobin Ni, Bryan Parno, Jonathan Protzenko, Tahina Ramananandro, Aseem Rastogi, Exequiel Rivas, Nikhil Swamy, Santiago Zanella-Béguelin, *ACM Transactions on Programming Languages and Systems (TOPLAS)*, 2026

Charon: An Analysis Framework for Rust, Son Ho, Guillaume Boisseau, Lucas Franceschino, Yoann Prak, Aymeric Fromherz, Jonathan Protzenko, *International Conference on Computer-Aided Verification (CAV)*, 2025

CUTECat: Concolic Execution for Computational Law, Pierre Goutagny, Aymeric Fromherz, Raphaël Monat, *European Symposium on Programming (ESOP)*, 2025, **Distinguished Artifact Award**

StarMalloc: Verifying a Modern, Hardened Memory Allocator, Antonin Reitz, Aymeric Fromherz, Jonathan Protzenko, *International Conference on Object-Oriented Programming, Systems, Languages and Applications (OOPSLA)*, 2024

Sound Borrow-Checking for Rust via Symbolic Semantics, Son Ho, Aymeric Fromherz, Jonathan Protzenko, *International Conference on Functional Programming (ICFP)*, 2024

Formalizing Date Arithmetic and Statically Detecting Ambiguities for the Law, Raphael Monat, Aymeric Fromherz, Denis Merigoux, *European Symposium on Programming (ESOP)*, 2024, **Best Tool Paper Award**

Modularity, Code Specialization, and Zero-Cost Abstractions for Program Verification, Son Ho, Aymeric Fromherz, Jonathan Protzenko, *International Conference on Functional Programming (ICFP)*, 2023

FastVer2: A Provably Correct Monitor for Concurrent, Key-Value Stores, Arvind Arasu, Tahina Ramananandro, Aseem Rastogi, Nikhil Swamy, Aymeric Fromherz, Kesha Hietala, Bryan Parno, Ravi Ramamurthy, *International Conference on Certified Programs and Proofs (CPP)*, 2023

Self-Repairing Neural Networks: Provable Safety for Deep Networks via Dynamic Repair, Klas Leino, Aymeric Fromherz, Ravi Mangal, Matt Fredrikson, Bryan Parno, Corina Pasareanu, *Workshop on Formal Methods for ML-Enabled Autonomous Systems (FoMLAS)*, 2022

Turning Catala into a Proof Platform for the Law, Alain Delaët, Denis Merigoux, Aymeric Fromherz, *Programming Languages and the Law (ProLaLa)*, 2022

Steel: Proof-oriented Programming in a Dependently Typed Concurrent Separation Logic, Aymeric Fromherz, Aseem Rastogi, Nikhil Swamy, Sydney Gibson, Guido Martínez, Denis Merigoux, Tahina Ramananandro, *International Conference on Functional Programming (ICFP)*, 2021

Fast Geometric Projections for Local Robustness Certification, Aymeric Fromherz, Klas Leino, Matt Fredrikson, Bryan Parno, Corina Pasareanu, *International Conference on Learning Representations (ICLR)*, *Spotlight Paper*, 2021

HACLxN: Verified Generic SIMD Crypto (for All Your Favourite Platforms), Marina Polubelova, Karthikeyan Bhargavan, Jonathan Protzenko, Benjamin Beurdouche, Aymeric Fromherz, Natalia Kulatova, Santiago Zanella-Béguelin, *ACM Conference on Computer and Communications Security (CCS)*, 2020

SteelCore: An Extensible Concurrent Separation Logic for Effectful Dependently Typed Programs, Nikhil Swamy, Aseem Rastogi, Aymeric Fromherz, Denis Merigoux, Danel Ahman, Guido Martinez, *International Conference on Functional Programming (ICFP)*, 2020

Steel: Scaling up Memory Reasoning for F*, Aymeric Fromherz, Denis Merigoux, *Automated Deduction for Separation Logics (ADSL)*, 2020

EverCrypt: A Fast, Verified, Cross-Platform Cryptographic Provider, Jonathan Protzenko, Bryan Parno, Aymeric Fromherz, Chris Hawblitzel, Marina Polubelova, Karthikeyan Bhargavan, Benjamin Beurdouche, Joonwon Choi, Antoine Delignat-Lavaud, Cédric Fournet, Tahina Ramananandro, Aseem Rastogi, Nikhil Swamy, Christoph Wintersteiger, and Santiago Zanella-Béguelin, *IEEE Symposium on Security and Privacy (Oakland)*, 2020

Symbolic Pathfinder for SV-COMP - (Competition Contribution), Yannic Noller, Corina S. Pasareanu, Aymeric Fromherz, Xuan-Bach D. Le, Willem Visser, *International Conference on Tools*

A Verified, Efficient Embedding of a Verifiable Assembly Language, Aymeric Fromherz, Nick Giannarakis, Chris Hawblitzel, Bryan Parno, Aseem Rastogi, and Nikhil Swamy, *Symposium on Principles of Programming Languages (POPL)*, 2019

Static Value Analysis of Python Programs by Abstract Interpretation, Aymeric Fromherz, Abdelraouf Ouadjaout, Antoine Miné, *NASA Formal Methods Symposium (NFM)*, 2018

Symbolic Arrays in Symbolic Pathfinder, Aymeric Fromherz, Kasper S. Luckow, Corina S. Pasareanu, *Java PathFinder Workshop*, 2016

Technical Reports

Layered Indexed Effects. Foundations and Applications of Effectful Dependently Typed Programming, Aseem Rastogi, Guido Martínez, Aymeric Fromherz, Tahina Ramananandro, Nikhil Swamy

Students

Antonin Reitz, PhD Student, November 2021-December 2025. Thesis defended: *Formal Verification and Evaluation of Memory Allocators*. Co-advised with Bruno Blanchet

Pierre Goutagny, PhD Student, September 2024-now. PhD topic: *Automated Verification of Catala Programs*. Co-advised with Raphaël Monat and Patrick Baillot.

Marco Bertoni, PhD Student, December 2025-now. PhD topic: *Automated and Incremental Side-Channel Analysis*. Co-advised with Yanis Sellami and Bruno Blanchet.

Florian Chivé, PhD Student, October 2026-now. PhD topic: *Interactive Program Verification Meets Abstract Interpretation*. Co-advised with Benoit Montagu and Bruno Blanchet

Teaching Experience

Fall-Winter 2026 **Proofs of Security Protocols**, SECURE, *Master Parisien de Recherche en Informatique (MPRI)*, Lecturer

Fall-Winter 2025 **Proofs of Security Protocols**, 2-30, *Master Parisien de Recherche en Informatique (MPRI)*, Lecturer

Fall-Winter 2024 **Proofs of Security Protocols**, 2-30, *Master Parisien de Recherche en Informatique (MPRI)*, Lecturer

Fall-Winter 2023 **Proofs of Security Protocols**, 2-30, *Master Parisien de Recherche en Informatique (MPRI)*, Lecturer

Spring 2019 **Secure Software Systems**, 18-732, *Carnegie Mellon University*, Head Teaching Assistant

Spring 2018 **Secure Software Systems**, 18-732, *Carnegie Mellon University*, Head Teaching Assistant

Professional Service

Program Committee member. ESOP 27, S&P 27, VSTTE 26, CPP 26, PriSC 26, CSF 26, ITP 25, USENIX Security' 25, CSF 25, POPL 24, USENIX Security' 24, JFLA' 23, USENIX Security' 23, USENIX Security' 22

Artifact Evaluation Committee member. ICFP' 22, ICFP' 21, POPL'21, ICFP'20, ISSTA'20

External Reviewer. ESOP'21, POPL'20, CPP'20, ESOP'20, FSE'19

Organizer of the ICFP Programming Contest. 2023, 2024